

Vitreoretinal tamponade and pressure complications

Part 2 | Tamponade choice and safety

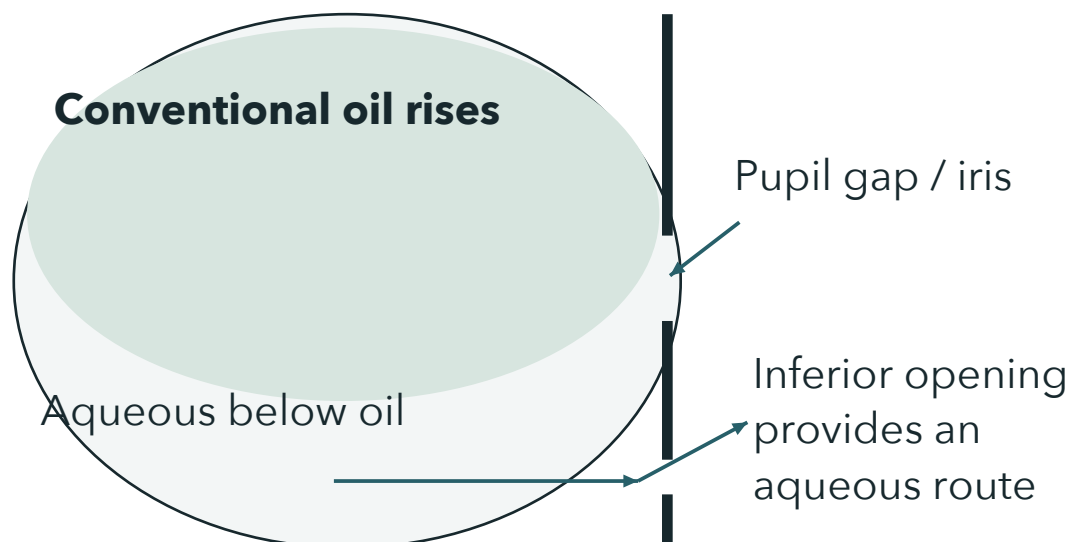
The chosen material changes follow-up and systemic precautions.

Finding	Implication
Gas	Temporary support; expansion and pressure risks
Conventional silicone oil	Buoyant long-term support requiring removal discussion
Gas present	Avoid nitrous oxide and unsafe altitude change
Lens and anterior route	Assess access to pupil and cornea

A tamponade is selected for the break pattern, retinal complexity and patient's ability to position and return. Explain how it supports the break and what later care it requires. Do not assume oil simply replaces a gas bubble with no further consequences: pressure, lens, corneal contact and removal become continuing decisions.

Acute oil-related pupillary block

Aqueous needs a route beneath a buoyant oil bubble.



Original aphakic section: an inferior iris opening provides an aqueous bypass.

In an aphakic eye, oil at the pupil can obstruct forward aqueous flow. A closed inferior iridotomy or iridectomy is therefore a structural explanation for a sudden pressure rise. Examine the actual route and restore it when indicated; pressure suppression alone does not correct a blocked passage. Phakic status does not exclude block when the diaphragm is compromised.

A patent opening does not exclude oil glaucoma

Locate the obstruction before selecting further treatment.

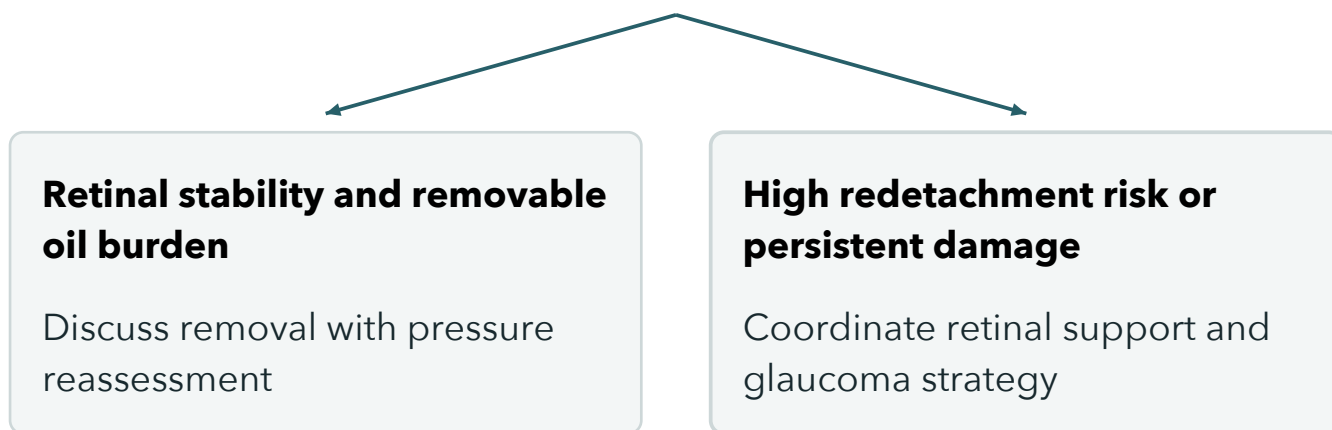
Finding	Implication
Acute closed passage	Pupillary-block mechanism
Open angle with droplets	Emulsified oil and outflow obstruction
Inflammation or steroid exposure	Additional pressure mechanisms
Corneal contact	Separate endothelial threat

Record angle anatomy, oil distribution, inflammation, medication exposure and optic-nerve reserve. The word postoperative is too broad to choose a treatment. A patent inferior opening addresses one mechanism; it cannot prevent every open-angle consequence of emulsification. Match the intervention to the structure that remains abnormal.

Oil removal is a benefit-risk decision

Removing oil may not reverse established outflow damage.

Would removal safely improve this eye?



Clinical series give differing pressure outcomes, so removal should not be presented as a guaranteed cure. Consider emulsification, duration, peripheral retinal stability and damage to drainage tissues. Explain both potential pressure benefit and the possibility of further glaucoma treatment, alongside the risk that removing support may destabilise the retina.

Cornea and vision after complex repair

Attachment alone does not explain all postoperative blur.

Finding	Implication
Anterior oil contact	Endothelial damage and keratopathy
Lens change	Optical cause of reduced acuity
Macular structural damage	Retinal limit despite attachment
High pressure	Additional optic-nerve threat

Reassess the optical pathway as well as the retinal result. A patient with an attached retina and worsening blur needs corneal, lens, pressure and macular review. State which component is reversible and what treatment could reasonably improve it. Avoid promising normal vision from another retinal procedure when the limiting tissue lies elsewhere.